Diego Adrian Ramirez Salazar

Robotics, Control & AI Engineer — Turbine Controls, Industrial Automation and Applied AI
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Profile

Robotics engineer (UDEM) working on turbine controls at power plants in Mexico and the U.S.: **GE Mark V/VI/VIe**, a Mitsubishi-to-**Mark VIe (TMR)** migration (∼540 instruments), SCADA and industrial networks (DNP3, Modbus). Automates field work with Python and applies PID/fuzzy control and machine learning (TensorFlow, PyTorch).

Professional Experience

PESS – Power Engineering Services and Solutions Apr 2025 – Present
Project Engineer *Power generation plants, Mexico / U.S.*

- Upgraded and maintained **GE Mark V, VI and VIe** turbine controls and GE/Siemens PLCs; tags and HMI screens in CIMPLICITY with historian; 4-20 mA/HART instrumentation.
- **International project (U.S., May 2025):** on-site robotic generator inspection at Lake Road Generating Plant (840 MW combined cycle, Connecticut), with **ELCID** stator-core test and winding inspection; equipment confirmed in good condition.
- Maintained **RTUs** linked to the grid operator (DNP3, Modbus, TCP/IP); with **Wireshark** and Python isolated periodic DNP3 packets (8 s) delayed by 30+ s and proved the source was external to the plant, **ruling out a plant fault**.
- Engineering, assembly and wiring of control cabinets and low-voltage MCCs; scale models in Visio.
- Automated thin-client setup in Python: **-70% configuration time** and uniform machines with no manual errors.

Technical work reporting & client contact *– Presidente Juárez combined-cycle plant, Rosarito, B.C.* *Nov 2026*

- **Main client contact:** daily progress reports, activity and labor-hour logs, and safety inspections of work areas.

Mitsubishi → GE Mark VIe (TMR) migration *– Cerro Prieto U10, Mexicali* *Sep – Nov 2026*

- Designed pre/post-migration connection verification checklists from plant engineering, **cutting ∼50% of the time** for loop-checking, disconnection and wiring vs. the previous project (team of 7).
- Structured plant engineering into an Excel database of **∼540 instruments** (description, location, card, terminals), the project's single source of truth.
- Documented the as-found state, loop-checked and labeled signals; disconnected the legacy system and wired the Mark VIe and its TMR network.
- Researched and proposed mitigations for the corrosive environment (thin clients and H_2S neutralization) to maximize equipment availability; **proposal approved and implemented**.
- Ran functional tests in GE ToolboxST, simulating system operation before startup.

Technical Skills

Industrial control: GE Mark V/VI/VIe (TMR), ToolboxST, PLCs (GE, Siemens), CIMPLICITY, Historian, DNP3, Modbus, TCP/IP, Wireshark, 4-20 mA/HART, ELCID
Control and robotics: PID / fuzzy control, kinematics, SLAM, Universal Robots e-Series, MATLAB/Simulink
AI / Machine Learning: TensorFlow, PyTorch, Keras, Scikit-Learn, TensorFlow.js, deep learning, computer vision (OpenCV)
Programming: Python (advanced), C++, Java, JavaScript
Embedded and other: Raspberry Pi, Tiva C, Linux, Git, Excel, Visio, NX, Fusion 360

AI and Robotics Projects

Digit recognizer – CNN in the browser | *Keras → TensorFlow.js, runs fully client-side* | live demo | code *2026*

- Convolutional network trained on MNIST (11,436 parameters, ∼45 KB); classifies hand-drawn digits in real time.

Image classification with neural networks | *AI Specialization, UDEM*

- Classical ML (Scikit-Learn) vs. deep learning (TensorFlow, PyTorch, Keras), with OpenCV preprocessing.

Autonomous navigation and SLAM | *Advanced Robotics, UDEM*

- Implemented simultaneous localization and mapping (SLAM) for autonomous navigation of a mobile robot.

4-axis rotary robot arm (R,R,R,R) *Aug – Nov 2023*

- Forward/inverse kinematics and control in Simulink; designed in Fusion 360, 3D-printed, brushless motors.

Education

Universidad de Monterrey (UDEM) San Pedro Garza García, N.L., Mexico
B.Eng. in Electronic Technology and Robotics *2018 – 2024*

Certifications and Languages

Industrial networks: Industrial Communication Protocols & Network Management (Vester Industrial Training Center, 2026) | Modbus Communication Protocol (Udemy, 2026)
STPS DC-3 safety (2025, renewed 2026): work at heights, confined spaces, electrical safety, lockout/tagout (LOTO), hazardous materials
AI and robotics: AI Specialization (UDEM) | Universal Robots e-Series Core Track | MATLAB (2021)
Languages: Spanish (native) | English (advanced, C1)